

KrishiSetu

Connecting Farmers with Agricultural Labor

PRODUCT REQUIREMENTS DOCUMENT (PRD) | v1.0 | 2025

Feature Name	Krishi Setu ("Farm Bridge" in Sanskrit)
Document Version	v1.0 — Initial Prototype Scope
Document Owner	Founder / Product Owner
Prepared For	Google AntigraVity / Development Team
Platform	Web Application (Mobile-Responsive)
Languages Supported	English, Hindi, Kannada
Target Geography	Karnataka, India (Prototype) — expandable pan-India
Status	DRAFT — Ready for Development Handoff

1. Executive Summary

KrishiSetu is a two-sided agricultural labor marketplace that directly connects farmers with skilled and unskilled agricultural laborers. The platform removes the chaos of informal hiring by giving laborers a clean digital identity (auto-generated resume) and giving farmers an instant view of nearby, available workers filtered by skill and timing.

The core insight is simple: a farmer in Haveri should not have to make 20 phone calls to find an irrigation worker available this weekend. And a laborer in Dharwad should not miss work opportunities because he has no way to advertise his skills. KrishiSetu solves both sides of this problem.

Problem	KrishiSetu Solution
Farmers can't find local labor quickly	Location-based labor discovery on tap
Laborers are invisible — no digital presence	AI auto-generates a structured resume from their inputs
Laborers are not literate in English/tech	Full Kannada / Hindi / English UI support
No standardized way to compare workers	Uniform resume template across all laborers

2. MUST-HAVE Features (Non-Negotiable)

The following features are the core of this product. Everything else is an enhancement. These MUST be delivered in the prototype. Do not compromise or descope any item in this section.

⚠ MUST-HAVE: Landing Page — Role Selection Gate

- ✓ When any user opens krishisetu section in the app, the FIRST thing they see is a role selection screen — no exceptions.
- ✓ Two clearly visible buttons: 'I am a Farmer' and 'I am a Labor'.
- ✓ No bypass — the user MUST tap one to proceed. This is the entry gate to both sides of the platform.

⚠ MUST-HAVE: Language Selection for Labor (Before Any Form)

- ✓ Immediately after a laborer taps 'I am a Labor', a language selection screen MUST appear.
- ✓ Three options: Kannada (ಕನ್ನಡ), Hindi (हिन्दी), English.
- ✓ Once a language is selected, EVERY piece of text on screen switches to that language — all questions, labels, buttons, error messages.
- ✓ No English fallback silently appearing. The entire UI must be in the chosen language. And should ask these specific questions to labour for his resume filling
1.name 2.contact number (must) 3.location of his(auto accees/type manually left to his wish)
4.wage/salary he wants 5.his availability 6.any special skills if he posses ,wt kind of work he can do

⚠ MUST-HAVE: Location-Based Labor Discovery for Farmers (GPS Required)

- ✓ When a farmer opens the app and taps 'I am a Farmer', the app MUST request GPS/location permission & choose language like English/hindi/kannada,even resume of labour should be shown in the language which farmer as opted
- ✓ The farmer sees ONLY laborers who are geographically near them — not a random nationwide list.
- ✓ Proximity filter must be configurable (e.g., within 5 km, 10 km, 25 km).
- ✓ This feature is compulsory. Without location, the farmer view should not show any profiles — only a message to enable location.

⚠ MUST-HAVE: AI Resume Auto-Generation for Labor Profiles

- ✓ When a laborer submits their details, an AI agent MUST automatically convert those raw inputs into a clean, structured resume.
- ✓ The resume template must be IDENTICAL for every laborer — same layout, same sections, same format.
- ✓ The farmer views polished resume cards — not raw form data.
- ✓ The AI must intelligently fill in the resume even if inputs are in Kannada or Hindi.
// For current scenario use dummy data to fill for labour as app is still in testing //

3. Product Flows & Screen Specifications

3.1 section of an app — The Entry Gate

name: **Krishi Setu**

Screen layout:

- Full-screen hero image: a farmland photograph with a soft dark overlay
- Centered logo and tagline: 'KrishiSetu —   ' (Your Farm's Bridge)
- Two large, prominent cards / buttons — side by side on desktop, stacked on mobile:
 -  I am a FARMER (Green, primary color)
 -  I am a LABOR (Earthy brown/amber, secondary color)
- Language toggle in top-right corner for the landing page itself (EN /  / )

3.2 Labor Flow — Complete Journey

Step 1 — Language Selection Screen

Shown immediately upon tapping 'I am a Labor'. Cannot be skipped.

- Headline: 'Please select your preferred language'
- Three large buttons with native script + flag emoji:
 -   (Kannada)
 -   (Hindi)
 -  English
- All subsequent screens render in the selected language from this point onward.

Step 2 — Labor Registration Form

All fields and labels are displayed in the selected language. The form is conversational and simple — no complex layout.

Field	Description	Required?
Full Name	Text input — first and last name	YES
Mobile Number	10-digit number with OTP verification	YES
Village / Town	Where the laborer is currently located	YES
Age	Numeric input	YES
Gender	Dropdown: Male / Female / Prefer not to say	YES
Primary Skill	Dropdown (see skill taxonomy in Section 3.3)	YES
Secondary Skill	Optional second skill from same list	NO
Availability From	Date picker — start of availability window	YES
Availability To	Date picker — end of availability window	YES
Time Preference	Checkboxes: Day shift / Night shift / Both	YES
Daily Wage (₹)	Numeric — self-declared, labeled as 'Negotiable'	YES

Years of Experience	Numeric input (0 = fresher is acceptable)	YES
Languages Spoken	Multi-select: Kannada, Hindi, Telugu, Tamil, English	YES
Photo (Optional)	Camera or file upload — adds trust on profile card	NO
Aadhaar Last 4 Digits	Optional — for verification badge on profile	NO

Step 3 — Review & Upload

- After all fields are filled, show a PREVIEW of what the profile will look like
- A single prominent button: 'Upload My Profile' / '□□□□□□□□ □□□□□□ □□□□' / '□□□□□□□□ □□□□□ □□□□'
- On tap, the AI agent is triggered — a loading screen shows: 'We are building your profile card...'

Step 4 — AI Resume Generation (Automated — No User Action Required)

This is the AI agent built inside the platform. Once the laborer taps Upload:

- An AI model receives the raw form data (in any language)
- It generates a standardized, professional Profile Card / Resume
- All text in the generated resume is output in the language preferred by the farmer regardless of input language (so farmers can read it in kannada /hindi/english)
- The laborer then sees their completed card with a success message

Resume Template — Fixed Sections (same for every laborer):

- Header: Name, Photo (if provided), Location Badge, Verified Badge (if Aadhaar provided)
- Skills: Primary Skill chip + Secondary Skill chip
- Availability: Date range + Shift preference (Day / Night / Both)
- Wage: '₹XXX/day (Negotiable)'
- Experience: X years
- Languages: Spoken language chips
- Contact: Tap-to-call button (visible only to authenticated farmers)

3.3 Agricultural Skill Taxonomy

The following skill options must be available in the labor registration dropdown — translated in all three languages in the UI:

#	Skill (English)	□□□□□	□□□□□□
1	Irrigation & Water Management	□□□□□□□ □□□□□□□□	□□□□□□ □□□□□□□
2	Sowing & Transplanting	□□□□□□□ & □□□□	□□□□□ □□ □□□□□
3	Harvesting (Crops)	□□□□□	□□□ □□□□
4	Pesticide & Fertilizer Application	□□□□□□□□□□ / □□□□□□□ □□□□□□□	□□□□□□□□ / □□□□□□□ □□□□□□□

5	Tractor / Farm Machinery Operation	□□□□□□□□□□ □□□□□	□□□□□□□□ / □□□□□□□
6	Land Preparation & Ploughing	□□ □□□□□□□ & □□□□□	□□□□ □□□□□□ □□□□□
7	Weeding	□□□ □□□□□□□□□□	□□□□□-□□□□□□□
8	Greenhouse / Polyhouse Work	□□□□□□ □□□□ □□□□	□□□□□□□□□□ □□□
9	Horticulture (Fruits & Vegetables)	□□□□□□□□□□	□□□□□□□□
10	Sericulture (Silk / Mulberry)	□□□□□□□ □□□□	□□□□ □□□□□□□
11	Poultry & Animal Husbandry	□□□□ / □□□ □□□□□□□	□□□□□□□□
12	Post-Harvest & Storage	□□□□□ □□□□ □□□□□□□□	□□□□ □□ □□□ □□□□□□□
13	General Farm Labor	□□□□□□□ □□□□ □□□□	□□□□□□□□ □□□□□□□□

3.4 Farmer Flow — Complete Journey

Step 1 — Location Permission

- Immediately after tapping 'I am a Farmer', the app requests GPS location (browser geolocation API)
- If denied: show a message — 'Please enable location to find laborers near you. This is required.'
- No location = no profiles shown. This is by design.

Step 2 — Labor Discovery Feed

This is the main farmer screen — a card-feed of nearby available laborers:

- Default radius: 10 km. Farmer can adjust via a slider: 5 km / 10 km / 25 km / 50 km
- Filter bar at top: Filter by Skill | Filter by Availability | Filter by Shift | Sort by Distance / Wage
- Each card shows: Name, Primary Skill, Location (village + km away), Shift, Wage, Availability dates, 'View Profile' button
- Tap 'View Profile' to see the full AI-generated resume card
- Tap-to-call button on the full resume (only if farmer is verified — future feature, for prototype just show number)

Step 3 — Farmer Filters (Enhanced Discovery)

Filter Type	Options
Proximity Radius	5 km / 10 km / 25 km / 50 km — slider or segmented control
Skill	All skills from taxonomy (Section 3.3) as filter chips
Availability	Date range picker — show only who is available during this window
Shift	Day Only / Night Only / Both
Wage Range	₹ slider — min to max daily wage

Experience	0-1 yr / 1-3 yr / 3-5 yr / 5+ yr
Sort By	Nearest First / Lowest Wage / Most Experienced / Recently Joined

4. Recommended Additional Features (Added by Product Team)

The following features are strongly recommended to make KrishiSetu a competitive, trustworthy product. These are in addition to the must-haves above.

4.1 Trust & Verification

- Verified Badge: Laborer who provides Aadhaar last 4 digits gets a blue check badge on their card
- Ratings & Reviews: After a work engagement, farmer can rate the laborer (1–5 stars). Rating is shown on the profile card.
- 'Hired X Times' badge: Shows how many times a laborer has been hired through KrishiSetu — builds social proof
- Report a Profile: Farmer can flag a fake or abusive profile for admin review

4.2 Smart Notifications

- New Laborer Alert: Farmer gets an SMS/WhatsApp notification when a new laborer registers within their preferred radius + skill
- Availability Expiry Warning: Laborer is reminded 2 days before their listed availability window ends — to update their profile
- Work Request Alert: (Future) Farmer can send a work request; laborer gets notified

4.3 Seasonal Demand Planner

- Admin panel shows which crops are in harvest season month-by-month across Karnataka
- Farmers get a banner: 'Paddy harvest season starting in your district — post your labor requirement now'
- Laborers who are inactive get prompted: 'Sowing season is starting near you — update your availability'

4.4 Labor Requirement Posting (Reverse Flow)

- Farmers can also POST a requirement: 'I need 4 irrigation workers for 3 days starting Oct 12, near Gadag'
- Matching laborers in the area receive an SMS/WhatsApp alert about the posting
- This is the reverse of the current flow and makes the platform truly two-sided

4.5 WhatsApp Integration

- Since many rural users are more comfortable with WhatsApp than apps, allow laborers to register via a WhatsApp chatbot (Future Phase 2)
- Farmer can share a laborer's profile card as a WhatsApp image to ask for peer recommendations

4.6 Offline-Friendly Design

- The labor registration form must work even on 2G connections — no heavy images or animations during form fill
- Profile cards are cached locally so farmers can browse recently loaded profiles without internet

4.7 Admin Dashboard (Internal)

- Total registered laborers by district and skill — heatmap view
- Daily/weekly new registrations chart
- Flagged / reported profiles queue
- Ability to manually verify or remove profiles

5. Technical Requirements

5.1 AI Resume Generation — Technical Specification

The AI agent must perform the following pipeline:

- INPUT: Raw JSON object with all form fields (may include Kannada / Hindi text in name, village, etc.)
- TASK: Transliterate or translate non-English inputs to English where needed
- TASK: Compose a professional plain-English bio paragraph (2–3 sentences) from the structured data
- OUTPUT: A structured JSON that maps to the fixed resume template sections
- RENDER: Frontend renders the JSON into the visual resume card (same template every time)

Sample AI prompt to be used:

```
You are an agricultural labor profile assistant.  
Given the following raw form data from a farm laborer (may be in Kannada,  
Hindi, or English),  
generate a professional, clean profile resume in English.  
Keep it simple — this is read by a farmer, not a corporate recruiter.  
Output as JSON with keys: name, location, primarySkill, secondarySkill,  
availability,  
shiftPreference, dailyWage, experience, languages, bio.  
INPUT: {form data}
```

5.2 Location & Proximity Query

- When a farmer opens the app, browser geolocation API captures latitude/longitude

- Query: SELECT * FROM laborers WHERE ST_DWithin(location, ST_MakePoint(farmer_lng, farmer_lat), radius_meters) AND available = TRUE ORDER BY distance
- Laborer location is set at registration time (village/district pinned to coordinates via Google Maps API)
- Laborer location is NOT tracked in real-time — it is the registered home location

5.3 Multi-Language Implementation

- Use i18n framework (react-i18next or similar)
- Maintain three translation JSON files: en.json, hi.json, kn.json
- All form field labels, button text, error messages, and instructions must be in all three files
- Language selection persists in localStorage for the session

6. A Note from the Product Owner

This product is built for people who feed this country.

A farmer in Karnataka does not have the luxury of waiting 3 days to find workers. A laborer does not have the privilege of explaining his skills in English on a corporate platform. KrishiSetu must feel like it was built by someone who understands both.

Please keep the interface clean, honest, and fast. No unnecessary pop-ups. No dark patterns. No overwhelming onboarding. Every extra tap we add, we lose a laborer who doesn't have time for it.

□ **KrishiSetu** — □□□□ □□□□□ □□□□ □□□□□□□□ □□□□□□ □□□□□□ □
(The bridge between farmers and agricultural labor)